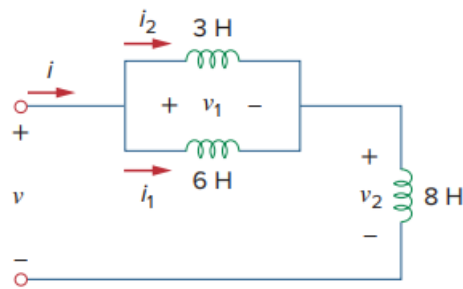


Problem

In the circuit of Fig. 6.34, $i_1(t) = 3e^{-2t}$ A. If $i(0) = 7$ A, find: (a) $i_2(0)$; (b) $i_2(t)$ and $i(t)$; (c) $v_1(t)$, $v_2(t)$, and $v(t)$.

Fig. 6.34:



Step-by-step solution

Step 1 of 7

Refer to the circuit in Figure 6.34 in the textbook.

(a)

By Kirchhoff's current law,

$$i = i_1 + i_2$$

$$i(0) = i_1(0) + i_2(0)$$

At $t = 0$,

$$i(0) = i_1(0) + i_2(0) \dots\dots (1)$$

The current i_1 is,

$$i_1(t) = 3e^{-2t} \text{ A}$$

At $t = 0$,

$$\begin{aligned} i_1(0) &= 3e^{-2 \times 0} \\ &= 3 \text{ A} \end{aligned}$$

Step 2 of 7

Calculate the current $i_2(0)$.

$$i(0) = i_1(0) + i_2(0)$$

$$i_2(0) = i(0) - i_1(0)$$

Substitute 7 A for $i(0)$ and 3 A for $i_1(0)$ in equation (1).

$$\begin{aligned} i_2(0) &= 7 \text{ A} - 3 \text{ A} \\ &= 4 \text{ A} \end{aligned}$$

Thus, the current $i_2(0)$ is 4 A.

Step 3 of 7

(b)

Calculate the voltage across 6 H inductor.

$$v_1(t) = 6 \frac{di_1(t)}{dt}$$

Substitute $3e^{-2t}$ A for $i_1(t)$.

$$\begin{aligned} v_1(t) &= 6 \frac{d}{dt}(3e^{-2t}) \\ &= 18(-2)e^{-2t} \\ &= -36e^{-2t} \text{ V} \end{aligned}$$

Since the 3 H inductor is parallel with 6 H inductor, the voltage across the both the inductor is same.

Calculate the current through the 3 H inductor.

$$i_2(t) = \frac{1}{L} \int_0^t v_1(t) dt + i_2(0)$$

Substitute $-36e^{-2t}$ V for $v_1(t)$ 3 H for L , and 4 A for $i_2(0)$.

$$\begin{aligned} i_2(t) &= \frac{1}{3} \int_0^t -36e^{-2t} dt + 4 \\ &= \frac{-36}{-6} e^{-2t} \Big|_0^t + 4 \\ &= 6(e^{-2t} - 1) + 4 \\ &= (-2 + 6e^{-2t}) \text{ A} \end{aligned}$$

Thus, the current $i_2(t)$ is $\boxed{(-2 + 6e^{-2t}) \text{ A}}$.

Step 4 of 7

By Kirchhoff's current law,

$$i(t) = i_1(t) + i_2(t)$$

Substitute $3e^{-2t}$ A for $i_1(t)$ and $(-2 + 6e^{-2t})$ A for $i_2(t)$.

$$\begin{aligned} i(t) &= 3e^{-2t} - 2 + 6e^{-2t} \\ &= (-2 + 9e^{-2t}) \text{ A} \end{aligned}$$

Thus, the current $i(t)$ is $\boxed{(-2 + 9e^{-2t}) \text{ A}}$.

Step 5 of 7

(c)

Calculate the voltage across 6 H inductor.

$$v_1(t) = 6 \frac{di_1(t)}{dt}$$

Substitute $3e^{-2t}$ A for $i_1(t)$.

$$\begin{aligned} v_1(t) &= 6 \frac{d}{dt}(3e^{-2t}) \\ &= 18(-2)e^{-2t} \\ &= -36e^{-2t} \text{ V} \end{aligned}$$

Thus, the voltage $v_1(t)$ is $\boxed{-36e^{-2t} \text{ V}}$.

Step 6 of 7

Calculate the voltage across 8H inductor.

$$v_2(t) = 8 \frac{di(t)}{dt}$$

Substitute $(-2 + 9e^{-2t})$ A for $i(t)$.

$$\begin{aligned} v_2(t) &= 8 \frac{d}{dt}(-2 + 9e^{-2t}) \\ &= 8[9(-2)e^{-2t}] \\ &= -144e^{-2t} \text{ V} \end{aligned}$$

Thus, the voltage $v_2(t)$ is $\boxed{-144e^{-2t} \text{ V}}$

Step 7 of 7

By Kirchhoff's voltage law,

$$v(t) = v_1(t) + v_2(t)$$

Substitute $-36e^{-2t}$ V for $v_1(t)$ and $-144e^{-2t}$ V for $v_2(t)$.

$$\begin{aligned} v(t) &= -36e^{-2t} - 144e^{-2t} \\ &= -180e^{-2t} \text{ V} \end{aligned}$$

Thus, the voltage $v(t)$ is $\boxed{-180e^{-2t} \text{ V}}$.